

If AI uses a lot of power, how, we haven't had massive blackouts?

Posted by u/Goodginger • 0 points • 29 comments

Excuse the maybe crazy question, but I'm new to this. I'm hearing that when a user enters an AI prompt, it tends to use a lot of power. And AI chat bots are very new, as far as I can tell. I asked it questions all day everyday. I assume a lot of people are too. This must create a lot of traffic or volume, whatever you call it . Did we already have the power in place, or are we having blackouts I haven't heard about?

Comments

u/robogame_dev • 1 points • 2025-12-11 06:51 UTC

Because all data centers in the world use about 1% of total electric demand, of which AI is just 15% - so 0.15% of total demand.

And of projected demand growth - so new electric demand - 10% is data centers and 90% is transit (electric cars), construction, and manufacturing.

Let that sink in, 90% of projected growth in electric demand has nothing to do with computers at all...

A lot of journalists are scared of AI so they amplify any negative story they can, they'll say "it uses x million watts" but not contrast that with the xxx billion watts that, for example, air conditioning uses.

https://www.carbonbrief.org/ai-five-charts-that-put-data-centre-energy-use-and-emissions-into-context/?utm_source=perplexity

u/Once_Wise • 2 points • 2025-12-10 20:58 UTC

You have just learned an important lesson. Everything you hear on the "news" is designed to get you to watch it. So the best way to do that, is to exaggerate the hell out of it. So that is why we don't have massive blackouts. The same reason we don't have massive blackouts with all the EVs. The grid grows with demand, just as it has since the time of Edison. Stop watching and listening to all the crap you are presented and start thinking a bit for yourself what is real and what is just hyperbole to get eyeballs.

u/MissingBothCufflinks • 0 points • 2025-12-10 20:00 UTC

Throttling. Thats it

u/RobbexRobbex • 2 points • 2025-12-10 19:53 UTC

It isn't that much power. My computer can run LLMs and it doesn't burn the house to the ground. The real issue is that the systems we use to run AI, and computers, and networks aren't very good for the

environment. Making the computer systems in general more environmentally friendly would solve this.

AI isn't the issue.

u/Puzzleheaded_Note363 • 1 points • 2025-12-10 19:36 UTC

No idea

u/Puzzleheaded_Note363 • 2 points • 2025-12-10 19:36 UTC

I feel like it may be a bit of an exaggeration on consumption per question. I say this because when I utilize a local (no internet) version of LLama or DeepSeek, my gaming computer doesn't blow up trying to keep up with the power required for the request. Hmm.

u/peter303_ • 1 points • 2025-12-10 19:07 UTC

Data centers are like 2% of the national grid now. But talk is up to 10% of the grid. I personally think there is not enough capital to finance everyone's plans, so not all this will be built.

u/Hegemonikon138 • 1 points • 2025-12-10 18:32 UTC

Inference and training are two different things. I run a 120b model locally and I will use more power running a video game for an hour then I use in two weeks to a month of AI

u/PopeSalmon • 10 points • 2025-12-10 18:20 UTC

data centers are a very small slice of the total power usage

when they set up a data center, they have to figure out how to buy power for it ,, they reach serious agreements w/ power providers, they don't just plug in all the computers and see whether there happens to be enough power for them ,, they're setting them up places that have excess power, or they're starting old power plants back up to power them, or xAI just set up a bunch of methane turbines outside memphis to power a center--- this sucks for people living there, b/c they have to breathe emissions from those turbines, but the data center isn't just plugged into the existing tennessee power system and let's see what happens, it's powered by some turbines they quickly set up

u/Bannedwith1milKarma • 4 points • 2025-12-10 19:42 UTC

This is the answer.

They wouldn't be allowed to turn on if they didn't make arrangements for the power first.

u/bloody_thuesday • 1 points • 2025-12-10 18:13 UTC

Production must absolutely follow consumption in real time

The networks are in poor condition in the US but it's big enough not to have to deal with that with other countries. In Europe the situation is very different, network in good condition and agreement between countries for transfer.

u/mrroofuis • 2 points • 2025-12-10 18:07 UTC

If they keep expanding at an exponential rate, we will.

We've already had electricity rates go up all across the US due to data centers

u/Immediate_Song4279 • 2 points • 2025-12-10 18:01 UTC

I have to take a middle stance on this.

There is an energy cost, and that needs to be accounted for. However, we already accept the concept

of energy expenditure, often in tiers according to individual use, industry, science, etc. What we often see is people arguing from extremes. Datacenters are significant, particularly to their surrounding area, and that does present very real design, engineering, and planning considerations. We live in a world where impoverished countries buy toxic e-waste, and desperate people burn them to extract small amounts of value.

We simply cannot take an isolated view of a systematic detail such as this, and need to consider human impact, efficiency, and curb development to yield to system constraints beyond just a on/off framing.

u/bayruss • 3 points • 2025-12-10 17:52 UTC

Not at capacity yet but they plan to expand by maybe 10x which will become a strain.

The better question is why didn't America spend money on a infrastructure 10 years ago? It would mean building out water and electric infrastructure lowering cost for Americans and allowing manufacturing to be done in the US... Maybe oil companies didn't want the government investing in green technologies. Maybe capitalism was stunting growth. Elong Tusk's EV war inadvertently lowered cost of batteries making Solar power cheaper than Coal.

We should be making the worlds biggest investment in energy/water infrastructure making a deal with the AI giants to subsidize it and rake in trillions from taxes.

u/Used-Study928 • 1 points • 2025-12-10 17:47 UTC

You are partly right, it depends on what we mean by AI but today you have micro models that are not very demanding, and that is surely how this story will end... billions of micro models!

u/IntroductionSouth513 • 1 points • 2025-12-10 17:32 UTC

any day now!

u/[deleted] • 1 points • 2025-12-10 17:30 UTC

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